

REVIEW

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A comparative review of the adsorption efficiency and performance variances of metal oxide nanoparticles and pristine carbon nanotubes for Pb²⁺ sequestration

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Abstract

Lead (Pb²⁺) contamination in aquatic systems poses a major threat to human health and environmental safety. Conventional remediation techniques such as chemical precipitation and ion exchange are often constrained by high operational costs, poor efficiency at trace concentrations, and secondary pollution. Nanotechnology has introduced materials like carbon nanotubes (CNTs) and metal oxide nanoparticles (MONPs) with superior physicochemical reactivity for Pb²⁺ removal. This review highlights recent advances (2002–2025) in adsorption–desorption behaviour of Pb²⁺ on MONPs and pristine CNTs, focusing on synthesis, surface properties, adsorption mechanisms, reusability, and toxicity. Reported findings reveal that MONPs, including Fe₃O₄, ZnO, MgO, and CuO, exhibit outstanding adsorption capacities (up to 1980 mg g⁻¹ for MgONPs), surpassing those of CNTs which usually range between 10 and 250 mg g⁻¹ under comparable conditions. This performance is linked to the abundance of chemically active surface sites and electrostatic interactions, which dominate adsorption on MONPs. Desorption studies confirm both materials are reusable over multiple cycles, though MONPs demonstrate higher regeneration efficiency and lower toxicity. Overall, MONPs present more scalable and eco-friendly alternatives for Pb²⁺ remediation. Future research should refine regeneration strategies, assess environmental safety, and develop hybrid systems for sustainable water treatment.

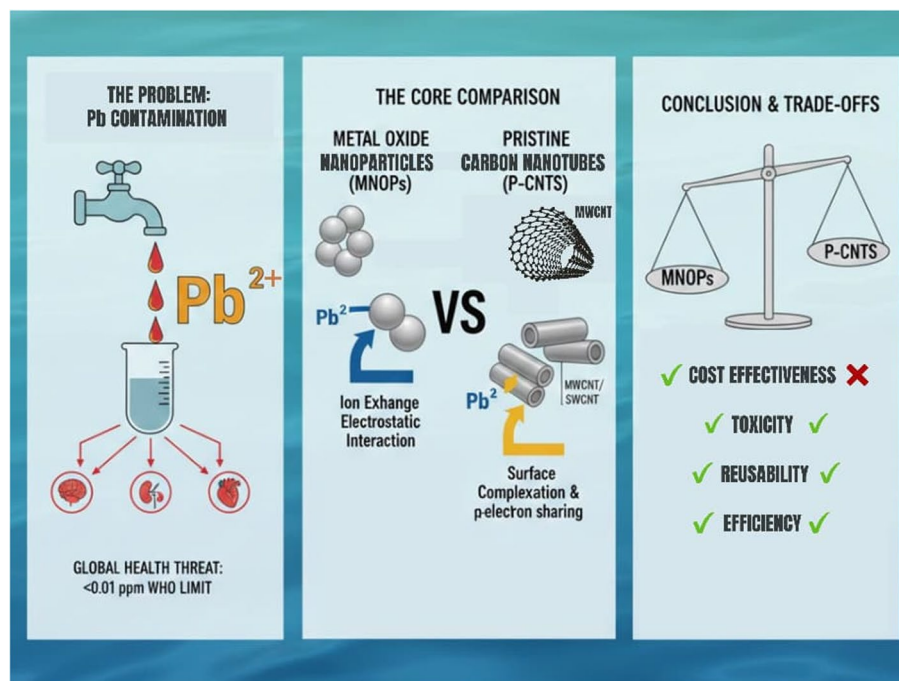
Keywords CNTs, MONPs, Adsorption, Desorption, Regeneration

1 Introduction

Heavy metals like Pb are metals and metalloids that have a specific gravity greater than 5 g/cm³. They are highly mobile, non-biodegradable, toxic, accumulative through food chains, and carcinogenic to humans [47, 58, 63, 86, 88]. They are generally hazardous to plant growth, human health, and animal safety [29, 30, 38, 126]. Other metals like mercury (Hg), nickel (Ni), arsenic (As), bismuth (Bi), chromium (Cr), cobalt (Co), and others are also classified as heavy metals [16].



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Graphical Abstract

Among these metals, lead poses serious concerns due to its toxicity, adverse effects, and no known biological function [37]. Lead poisoning is the term used for lead toxicity, which may be acute or chronic [19]. “Lead poisoning” has been investigated to cause encephalopathy, hepatitis, kidney damage, high blood pressure, abortion, low sperm count, anorexia nervosa, cancer (particularly gastric cancer), brain damage, malaise, anaemia, autism, dyslexia, and several damages to body organs [14, 51, 89]. In response to these risks, the World Health Organization has established a permissible limit of 0.01 ppm of lead for drinkable water [5, 98]. However, lead continues to find its way into water bodies through sources such as ammunition production, battery production, plastic stabilizer production, pigment processing, smelting, electroplating, alloys, cable sheathings, pesticide application, and use of leaded fuels, among others [38, 48, 64, 65, 91]. As a result, the development of efficient and sustainable treatment technology for the sequestration of lead from water remains an environmental priority.

Several water treatment techniques have been used to sequester lead from aqueous media, but these conventional techniques suffer the demerit of secondary sludge production, water wastage, high operational cost, slow kinetics, operational cost, and maintenance challenges. Liu et al., [67], Nwoke, [87], Vaneeckhaute, [124], Zhang, [140]. These shortcomings have inspired interest in adsorption-based water treatment technologies, especially those involving nanomaterials, due to their operational simplicity, high efficiency, and potential for regeneration [19]. Adsorption using nanoparticles provides advantages such as enhanced surface areas and enhanced active sites for binding. Nanoparticles (NPs) eliminate most of these problems as it is efficient, cost-effective, simple, easy to execute, and the adsorbents can be functionalized and recycled [107]. Some disadvantages have been associated with the use of NPs for adsorption procedures,

such as possible leaching of Nano-adsorbent into water, and production of secondary pollutants [93].

Among the different classes of nanoparticles used as adsorbents, metal oxide nanoparticles (MONPs) and carbon nanotubes (CNTs) stand out as two of the most widely researched for the sequestration of lead. MONPs have been reported to show strong affinity for metals, redox potential, and even magnetic properties, while CNTs have demonstrated high surface area and strong mechanical strength [92, 109]. Notable limitations such as agglomeration, instability, and potential toxicity have been reported for MONPs, while CNTs have been reported to suffer from agglomerations, disposal concerns, and regeneration challenges [5, 93, 101].

Numerous studies and reviews have separately examined the potency of MONPs and CNTs for lead sequestration; however, direct systematic comparison between pristine CNTs (P-CNTs) and MONPs with regard to lead sequestration remains limited and fragmented. Existing literature often focuses on adsorption capacity with little consideration for performance trade-offs such as regeneration efficiency, scalability, environmental safety, and practical deployment constraints.

Unlike previous reviews that consider MONPs and CNTs independently, this study provides a structured comparative evaluation based on adsorption capacity, kinetics, regeneration efficiency, scalability, and toxicity. By focusing exclusively on pristine P-CNTs, this study provides a baseline for assessing the performance of CNTs relative to MONPs without the influence of functionalization.

2 Methodology

This study focuses on the comparison of MONPs and pristine carbon nanotubes (P-CNTs) in adsorbing aqueous divalent lead ions. A structured protocol was adopted to guide the identification, screening, eligibility assessment, and adoption of literature as shown in Fig. 1.

2.1 Search strategy

A comprehensive literature search was performed across numerous databases, including Scopus, Web of Science, ScienceDirect, Google Scholar, and PubMed, and the search covered peer-reviewed publications from 2000 to 2025, which represents the period of increased interest in Nano-adsorbent research. Search strings formed with keywords such as “Pb²⁺ adsorption”, “lead removal”, “metal oxide nanoparticles”, “pristine carbon nanotubes”, “CNT adsorption”, “Nano-adsorbents”, and “water treatment” to assess relevant literature, and the reference list of related studies was manually screened to identify relevant studies.

2.2 Eligibility conditions

Screening of studies was done based on predefined conditions that relevant studies must investigate lead sequestration from aqueous media, employ MONPs or P-CNTs as primary adsorbent, report quantitative adsorption metrics such as adsorption capacity and removal efficiency, and be published in English peer-reviewed journals. Studies focusing on hybrid (MONPs-CNTs) and functionalized CNTs were excluded from the comparison, but were considered for contextual discussions where relevant. Materials lacking adequate experimental details were excluded.

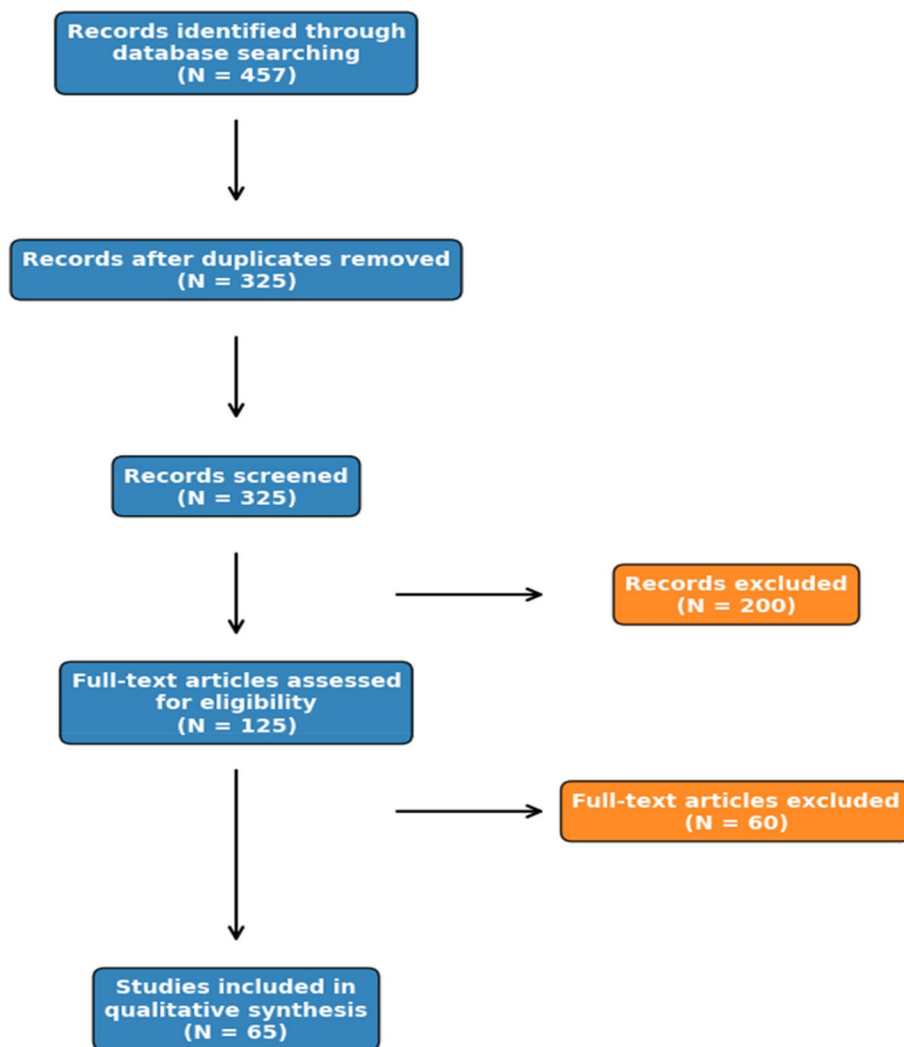


Fig. 1 PRISMA flow diagram

2.3 Study selection, data extraction, and quality assessment

Materials sourced from the databases were compiled, and duplicates were removed. The titles and abstracts were assessed to confirm relevance to the adsorption of lead using MONPs or CNTs from aqueous systems. Thereafter, studies meeting all the inclusion criteria were adopted in the comparative analysis.

Variables such as adsorbent type, synthetic method, particle size, experimental conditions, optimal adsorption capacity or percentage removal, isotherm, kinetics, and regeneration ability were extracted from the included studies. Studies employing standard adsorption protocols were considered to ensure data reliability. Variations resulting from differences in experimental conditions were acknowledged during the comparative assessment. Lastly, the comparative assessment involved evaluation of the mechanism of adsorption to infer the dominant mechanism, assessing performance indicators such as regeneration efficiency, material stability, agglomeration potential, and environmental friendliness to elucidate performance trade-offs between MONPs and P-CNTs. The findings were presented using thematic analysis and structured Tables to support evidence-based conclusions.

3 Metal oxide nanoparticles (MONPs)

A comprehensive review of literature has shown that MONPs are among the most studied Nano-adsorbents for the sequestration of heavy metals from aqueous solutions. Properties such as high surface area, redox activity, stability, and enhanced surface action have been reported in numerous studies as the primary factors responsible for their adsorption performance [14, 42].

A wide range of MONPs have been investigated for their potential for lead sequestration, such as CuONPs, FeONPs, ZnONPs, CaONPs, CaTiO₃NPs, Al₂O₃NPs, NiONPs, CeO₂NPs, TiO₂NPs, MnO₂NPs, MnFe₂O₄NPs, CoFe₂O₄NPs, SnO₂NPs, and MgONPs. A similar observation which has been observed across these MONPs is that adsorption behavior is strongly influenced by the density of surface hydroxyl groups and the material's point of zero charge as these factors control the electrostatic attraction and surface complexation with Pb ions [42, 111].

The choice of synthetic route (especially green synthesis/hydrothermal) has been reported to be the primary determinant of MONP properties, including surface area, crystallite size, pore size distribution, and available functional groups [20, 112].

3.1 Copper oxide nanoparticles (CuONPs)

The current study noted that CuONPs demonstrate moderate to high adsorption potential for lead (between the range of 84–153 mg/g), with the performance reportedly dependent on synthesis method, particle size, and experimental conditions. Green synthesized CuONPs were noted to show comparable or superior efficiency relative to the chemically synthesized CuONPs. Pseudo second order kinetics was consistently reported suggesting that chemisorption dominates Pb uptake, while isotherm modelling mostly align with Langmuir and Freundlich models, showing the coexistence of homogeneous and heterogeneous sites.

Zaidi et al., [136] synthesized CuONPs via chemical co-precipitation and reported 84.2% removal of Pb²⁺ from a 5 ppm solution in 120 min at a pH of 4.5. The adsorption process was best described by the Freundlich model ($R^2 = 0.922$), suggesting heterogeneous surface adsorption and a chemisorption mechanism. Mahmoud et al., [71] performed green synthesis using mint leaves and orange peel extract to produce CuONPs, both of which showed high adsorption capacities ($q_m = 88.8$ and 82.8 mg/g, respectively), under identical conditions. These findings highlight the effectiveness of bio-mediated synthetic approaches in producing functional adsorbents while reducing environmental impacts associated with conventional chemical synthesis.

Microwave-assisted synthesis has also been shown to yield efficient CuONPs for Pb sequestration. Farghali et al., [36] reported that CuONPs prepared using microwave radiation demonstrated adsorption behaviour consistent with the Langmuir adsorption isotherm ($R^2 = 0.996$), indicating monolayer adsorption on homogeneous active sites. The kinetics followed the pseudo-second order (0.999), suggesting that chemisorption was the major rate-limiting step.

The study by Hassan and Mahdi [45] further demonstrated the high removal efficiency and fast adsorption equilibrium of CuONPs. They reported effective Pb sequestration using CuONPs with a crystallite size of about 7.43 nm, and achieved equilibrium under optimized conditions (adsorbent dose of 0.1 g, pH of 3.5, initial lead concentration of 100 ppm, and temperature of 25 °C).

Collectively, these findings indicate that CuONPs exhibit strong affinity and fast adsorption kinetics for Pb, and that the synthetic route influences the sorption performance. Table 1 shows some reported studies on the sequestration of Pb using CuONPs.

3.2 Iron oxide nanoparticles (FeONPs)

Iron oxides occur in different forms such as maghemite ($\gamma\text{-Fe}_2\text{O}_3$), hematite ($\alpha\text{-Fe}_2\text{O}_3$), magnetite (Fe_3O_4), and goethite (FeOOH). These polymorphic structures provide the specific surface chemistries and adsorption behaviors relevant for remediation applications. FeONPs, especially magnetite are among the most extensively studied Nano-sorbents for the removal of heavy metals owing to their magnetic recoverability and strong adsorption performance [65, 77]. Recorded Pb removal efficiencies mostly exceed 90%, while adsorption capacities ranging from 50 to 170 mg/g were also noted in optimized systems.

Comparative studies reveal that the crystallographic phase is a major determinant of adsorption efficiencies, with magnetite generally outperforming hematite and other polymorphs. This variation highlights the differences in surface charge distribution, redox properties, and active site density [79]. Green-synthesized FeONPs were noted to exhibit superior performance in comparison to the conventionally produced counterparts, signifying the importance of surface functionalization and biogenic capping agents [64]. This is also supported by the prevalence of pseudo second order kinetics and Langmuir isotherm, signifying that the removal of using FeONPs is mediated by the homogenous coverage on surface chemical groups.

Rahimi et al., [94] compared hydrated lepidocrocite and goethite for Pb sequestration and reported that both materials demonstrated an optimum pH of 5.2, and followed Langmuir and Sips adsorption isotherms. However, goethite showed 1.64 times better adsorption capacity than lepidocrocite, showing the effect of mineral phase on adsorption performance.

Similarly, different crystalline forms of Fe_2O_3 NPs have shown varying efficiencies. Hexagonal Fe_2O_3 NPs with crystallite sizes ranging from 20 to 60 nm achieved a Pb removal efficiency of 96.73% within 60 min under alkaline pH [77]. Other studies report adsorption efficiencies of 95% and 92.47% using $\gamma\text{-Fe}_2\text{O}_3$ NPs [6, 103], while $\alpha\text{-Fe}_2\text{O}_3$ NPs have shown comparatively lower adsorption capacities [13]. These reports show that crystallographic structures greatly influence adsorption performance.

Table 1 Some reported studies on the sequestration of Pb using CuONPs

MONPs	I.C (ppm)	q_m (mg/g) or $R(\%)$	Isotherm	Ki-net-ics (R^2)	Dose (g)	Time (min)	pH	Temp ($^{\circ}\text{C}$)	Crys-tallite size (nm)	Sur-face area (m^2/g)	Reference
CuONPs	5	84.2%	Freundlich	PSO	1.0	120	4.5		21.6		Zaidi et al., [136]
CuONPs – 1	5	88.8 mg/g	Freundlich	PSO	0.33	60	6		-		Mahmoud et al., [71]
CuONP		153.8	Freundlich	PSO		180	7		-		Mashentseva et al., [76]
CuONPs (oval)	100–300		Langmuir	PSO	0.1	240	6.5		20	5.62	Farghali et al., [36]
CuONPs	100	84.1%	-		0.1	30	3.5	25	7.43		Hassan and Mahdi, [45]

Lingamdinne et al., [64] synthesized Fe_3O_4 NPs from tangerine extract and reported 99% adsorption efficiency after 90 min. The process followed Langmuir isotherm ($R^2 = 0.996$), and pseudo-second order (PSO) kinetics ($R^2 = 0.998$), signifying monolayer chemisorption mechanisms. Similarly, Bagbi et al., [20] co-precipitated Fe_3O_4 NPs and achieved almost complete removal of Pb^{2+} under optimized conditions of adsorbent dose of 0.2 g/L, and pH 5. Hassan and Mahdi, [45] reported 88% Pb removal using Fe_3O_4 NPs within 30 min under acidic conditions, while Sarkar and Sarkar [108] reported 90% removal at pH 6. Comparable adsorption capacities were also reported by Rehman et al., [99] and Karami [54], who recorded respective capacities of 96.77 mg/g and 112.9 mg/g.

Using the green synthesis approach, El-Kassas et al., [34] produced Fe_3O_4 NPs from marine algae *Padina pavonica* and *Sargassum acinarium*, and reported percentage adsorptions of 91% and 78%, respectively. Lung et al., [68] demonstrated the superiority of green-synthesized Fe_3O_4 NPs over the conventional counterpart by recording a removal percentage of 98.8% compared to 59.4% under ideal conditions.

Tamez et al., [118], in their work using Fe_3O_4 NPs and Fe_2O_3 NPs, noted respective adsorption efficiencies of 166.67 and 47.62 mg/g, highlighting the superior adsorption efficiency of magnetite. A major advantage of FeONPs, especially Fe_3O_4 NPs, is their magnetic properties, which enable rapid recovery using an external magnetic field. This feature enhances their suitability for water treatment as it reduces secondary contamination risk and improves reusability properties [20, 48, 60]. Fig. 2 depicts the sequestration of Pb using a magnet, while Table 2 shows some reported studies on the sequestration of Pb using FeONPs.

3.3 Zinc oxide nanoparticles

Reported experimental adsorption studies have repeatedly shown that particle size, synthetic route, and surface functionalization significantly affect the performance of ZnONPs in Pb removal, with adsorption capacities of as low as 15 mg/g and as high as >500 mg/g reported. Green-synthesized ZnONPs have been reported to exhibit superior performance over conventionally produced counterparts just as observed with

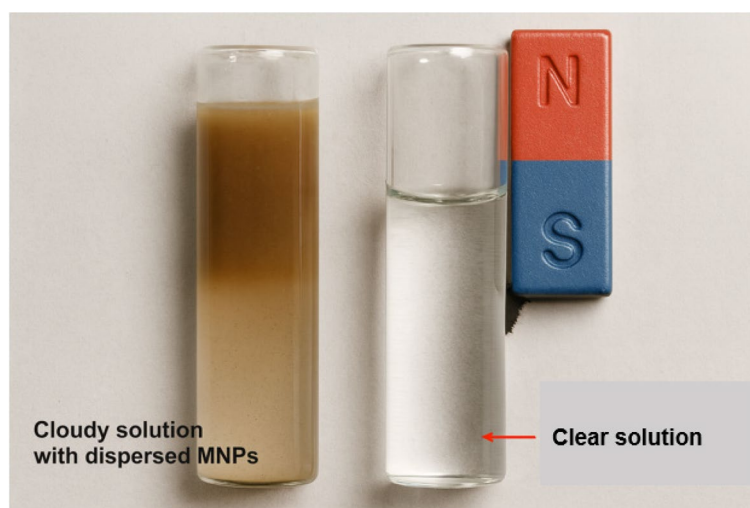


Fig. 2 Separating magnetite NPs using a simple magnet

Table 2 Some reported studies on the sequestration of Pb using FeONPs

MONPs	I.C (ppm)	q _m (mg/g) or R(%)	Iso-therm (R ²)	Kinetics (R ²)	Dose (g)	Time (min)	pH	Temp (°C)	Crys-tal-lite size (nm)	Sur-face area (m ² /g)	Refer-ence
Fe ₃ O ₄ NPs	100	88%			0.1	30	3.5	25	12.04		Hassan and Mahdi, [45]
Fe ₃ O ₄ NPs	100	90%			10	10			24		
Padina pavonica-Fe ₃ O ₄ NPs	75–175	91%				75	6	25			El-Kassas et al., [34]
Sargassum acinarium-Fe ₃ O ₄ NPs	75–175	78%				75	6	25			El-Kassas et al., [34]
Fe ₃ O ₄ NPs	100	90%				10	6	24			Sarkar and Sarkar, [108]
Water Soluble Fe ₃ O ₄ NPs	10	90%	Lang-muir (0.98)	PSO (0.998)		2			20		Wang et al., [131]
Fe ₂ O ₃ NPs	50	96.73%			0.01	60	8		20–60		Miri et al., [77]
T-Fe ₃ O ₄ NPs	10	99%	Lang-muir (0.9960)	PSO (0.9980)	0.6	90	5	25	200–300		Lingam-dinne et al., [64]
Fe ₃ O ₄ NPs	50	~ 100%		PSO (0.9938)	0.2	30	5	25	10.4	73.28	Bagbi et al., [20]
Fe ₃ O ₄ NPs	5	96.77 mg/g	Lang-muir (0.971)		0.05	180	5	40	22	95	Rehman et al., [99]
Fe ₃ O ₄ NPs	1000	98%	Lang-muir (0.975)	PSO (0.99)	0.05	90	6			46.8	Taqui et al., [119]
Fe ₃ O ₄ NPs	50	112.9 mg/g	Lang-muir (0.998)	PSO	0.05	20	5.5	25	55–65		Karami, [54]
Conven-tional Fe ₃ O ₄ NPs	20	59.4%	Lang-muir (0.993)	PSO (0.999)		120	5.5	25			Lung et al., [68]
Green syn-thesized Fe ₃ O ₄ NPs	20	98.8%	Freun-dlich (0.998)	PSO (0.999)		120	5.5	25			Lung et al., [68]
Fe ₃ O ₄ NPs-CTAB	10	19 mg/g		PSO		4,320	5	45	31.2	11.3	Kumari et al., [60]
Fe ₃ O ₄ NPs		34.9 mg/g	Lang-muir (0.98)	PSO (0.99)			5	45		12.7	Rajput et al., [96]
Fe ₃ O ₄ NPs powder	9.66	98%	Lang-muir (0.97)			40–100	6	38.25	73.05		Mahanty et al., [70]
Fe ₃ O ₄ NPs	0.03	166.67 mg/g	Lang-muir		0.01	20	4–6		19.9		Tamez et al., [118]
Fe ₂ O ₃ NPs	0.03	47.62 mg/g	Lang-muir		0.01	20	3–4		12.9		Tamez et al., [118]

Table 2 (continued)

MONPs	I.C (ppm)	q _m (mg/g) or R(%)	Iso-therm (R ²)	Kinetics (R ²)	Dose (g)	Time (min)	pH	Temp (°C)	Crys-tal-lite size (nm)	Sur-face area (m ² /g)	Refer-ence
α-Fe ₂ O ₃ NPs	20	27 mg/g	Lang-muir (0.989)	PSO(0.999)	0.05	120	5.5	25	36		Ali et al., [13]
γ-Fe ₂ O ₃ NPs	20–400	71.24 mg/g or 95%	Lang-muir	PSO	0.05	240	6	25	10–15	321.64	Roy and Bhat-tacharya, [103]
γ-Fe ₂ O ₃ NPs		92.47%				120	7.5	45	14		Ahmadi et al., [6]
Fe ₂ O ₃ @ Microalgae NPs	80	62.63 mg/g	Lang-muir (0.95)	PSO (0.999)	0.05	30	5			156	Shen et al., [114]
Mixed Fe- ₃ O ₄ NPs and α-Fe ₂ O ₃ NPs	2	617.3 mg/g or 96.24%				60	7	25	4–52		(M. A. Ahmed et al., [8]
α-FeOOH	5	53.88%	Lang-muir, Sips			105	5.2			145.6	Rahimi et al., [94]
γ-FeOOH	5	87.74%				155	5.2			268.8	Rahimi et al., [94]
FeOWNPs-Alg		264 mg/g	Freun-dlich	PSO		30	6		16–24	5.43	Mah-moud et al., [74]
O-Fe ₃ O ₄	25	99%	Lang-muir	PSO	2	30	5.5				Ghosh et al., [42], Li, [62]

CuONPs and FeONPs, highlighting the importance of surface functionalization and capturing agents in sequestration performance.

Azizi et al., [18] synthesized ZnONPs with a mean size of 10.01 nm through the thermal decomposition of zinc hydroxide and plant-based sesquiterpene, zerumbone, and recorded a Pb adsorption efficiency of 93%. Samad et al., [106] reported the use of silty clay to synthesize ZnONPs and observed a Pb adsorption capacity of 14.54 mg/g. On the other hand, ZnONPs stabilized with *tinospira cordifolia* gave an adsorption capacity of 506 mg/g for aqueous lead [128].

These studies underscore the importance of the green synthesis approach and surface functionalization techniques in optimizing ZnONPs for use in the sequestration of Pb.

3.4 Calcium oxide nanoparticles (CaONPs) and calcium titanate nanoparticles (CaTiNO₃NPs)

Calcium-based nanoparticles, mostly CaONPs and CaTiNO₃NPs, have reportedly shown potentials as the removal of Pb from aqueous systems, with their effectiveness linked to their high surface reactivity, ion-exchange capacity, and abundant active sites for adsorption.

Jalu et al., [51] applied the sol-gel technique for the synthesis of CaONPs from egg shells and reported a Pb adsorption efficiency of 99.07%.

CaTiO₃NPs have similarly been investigated with multiple sources reporting high Pb adsorption capacities. Zhang et al., [139] observed a Pb adsorption efficiency of 141.8 mg/g with CaTiO₃NPs, Hang et al., [44] reported an adsorption efficiency of 261.8 mg/g in their study on Pb adsorption, while Song et al., [116] recorded an adsorption efficiency of 197.96 mg/g.

3.5 Aluminum oxide nanoparticles (Al₂O₃NPs)

Aluminum oxide nanoparticles (Al₂O₃NPs), especially the gamma-phased alumina (γ-Al₂O₃NPs), have been widely studied for Pb sequestration applications because of their porosity, stability, and abundant surface hydroxyl groups [46]. Experimental studies have noted variations in Pb adsorption capacity depending on the synthetic method and experimental conditions.

Tabesh et al., [117] synthesized γ-Al₂O₃NPs using the sol-gel method and reported Pb adsorption efficiency of 97% under optimized conditions. Similarly, Rahmani et al., [95] reported an adsorption capacity of 125 mg/g, showing the high Pb sequestering potential of γ-Al₂O₃NPs from aqueous solutions.

However, Shokati Poursani et al., [115] reported an adsorption capacity of 6 mg/g in their study using γ-Al₂O₃NPs, demonstrating the role of surface chemistry and experimental conditions in determining the adsorption efficiency.

3.6 Other MONPs

Several less studied MONPs have also demonstrated varying degrees of effectiveness for Pb sequestration from aqueous systems. These are materials such as manganese oxide, nickel oxide, magnesium oxide, cerium oxide, cobalt ferrite, tin oxide, and titanium dioxide, which have shown distinct properties necessary to aid sequestration performance.

Manganese oxide nanoparticles (Mn₂ONPs) and composites demonstrate high adsorption capacities (between 121 and 260 mg/g) [24, 31, 73], whereas other counterparts such as Cerium oxide nanoparticles (CeO₂NPs) and Nickel oxide nanoparticles (NiONPs) have shown lower capacities of 5 mg/g and 56 mg/g respectively [1, 72, 129].

Cobalt ferrite nanoparticles have been reported to show moderate Pb adsorption potentials with reported removal capacities falling between 32 and 63 mg/g [11, 27].

Titanium dioxide and tin oxide nanoparticles (TiO₂NPs and SnO₂NPs) have also been reported to show good sorption properties, with respective capacities of 90.13 mg/g and 98.13 mg/g recorded [99]. Complete removal (100%) of Pb²⁺ was also documented by Engates and Shipley [35] using TiO₂NPs.

MgONPs stands out with exceptionally high adsorption capacity (up to 1980 mg/g and 94%) showing strong affinity for Pb [21, 28].

Several consistent trend was observed across the MONPs and they include:

1. Adsorption is mostly governed by electrostatic interaction and surface complexation
2. Pseudo second order kinetics dominate, showing chemisorption
3. Green synthesis enhances performance

Table 3 shows reported studies on the performance of ZnONPs, CaONPs, and Al₂O₃NPs in Pb removal, while Table 4 highlights the performance of other MONPs in the sequestration of Pb.

Table 3 Performance of some ZnONPs, CaONPs, and Al₂O₃NPs in the sequestration of Pb

MONPs	I.C (ppm)	q _m (mg/g) or R(%)	Isotherm (R ²)	Kinet-ics (R ²)	Dose (g)	Time (min)	pH	Temp (°C)	Crys-tallite size (nm)	Sur-face area (m ² /g)	Reference
ZnONP		93%	Langmuir	PSO	1	60	5	70	10.01		Azizi et al., [18]
ZnONP		163.8	Freundlich	PSO		240	7				Mashent-seva et al., [76]
SC-ZnONPs		14.54 mg/g			0.2	30	6	45			Samad et al., [106]
ZnO@TnCNPs	100	506 mg/g	Langmuir	PSO	0.01	360	6.8		295		Vyas et al., [128]
CaONPs	75.46	99.07%	Langmuir (0.9963)	PSO (0.9982)	0.838		6.9		24.34	77.4	Jalu et al., [51]
CaTiO ₃ NPs		141.8 mg/g	Freundlich (0.993)		0.1	40	6	25			Zhang et al., [139]
CaTiO ₃ NPs	750	261.8 mg/g	Langmuir (0.999)	PSO (0.9954)	1	60	6	20			Hang et al., [44]
CaTiO ₃ NPs	200	197.96 mg/g	Langmuir (0.999)	PSO (0.993)	0.05	30	5	25	27	359.5	Song et al., [116]
γ-Al ₂ O ₃ NPs	50	97%	Freundlich (0.9955)	PSO (0.9959)	0.01	20	5		6–13		Tabesh et al., [117]
γ-Al ₂ O ₃ NPs	25–150	125 mg/g	Langmuir (0.995)	PSO (0.999)	0.4	180	4	25	18.43–21.04	206	Rahmani et al., [95]
γ-Al ₂ O ₃ NPs		6 mg/g	Langmuir (0.996)	PSO (0.978)	0.03	240		25	11.5	125.4	Shokati Poursani et al., [115]

4 Carbon nanotubes (CNTs)

Cho et al., [26] described CNTs as allotropes of carbon possessing high adsorption capacity as a result of their layered and hollow structure, p electron-sharing ability, and large surface area. CNTs can best be visualized as a rolled sheet of graphene possessing an axial symmetry [26]. CNTs are divided into two main categories depending on the number of graphene layers rolled into a coaxial array, and as such, we have the Single-Walled Carbon Nanotubes (SWCNTs) and the Multi-Walled Carbon Nanotubes (MWCNTs). SWCNTs possess diameters within the range 0.4 to 20 nm and lengths ranging from 100 nm to 10 μm, with the lengths and diameters depending on the synthetic method. In MWCNTs, the respective tubes are separated by 0.34 to 0.36 nm. MWCNTs typically possess diameters ranging from 1 nm to 300 nm and lengths from 1 μm to 150 μm [26].

CNTs are chemically inert, have a high surface area, and possess water transport properties [92]. They have found useful applications as: adsorbents, delivery for genes and seeds in agriculture, antimicrobial agents against pathogens, and Nano-sensors [104, 122]. Mubarak et al., [82] also reported CNTs as having large surface areas for the removal of lead and other heavy metal ions. As adsorbents, CNTs have the general advantages of high adsorption capacity, good thermal and chemical stabilities, low mass density, high porosity, and eco-friendliness [41].

The major methods used in the synthesis of CNTs are carbon arc discharge, laser ablation, and chemical vapour deposition (CVD) techniques. The arc discharge method uses high temperature (> 1700 °C), low pressure, and expensive gas to synthesize CNTs with minimum structural defects. Laser ablation uses a vacuum with controlled inert gases to

Table 4 Performance of other MONPs in the sequestration of Pb

MONPs	I.C (ppm)	q _m (mg/g) or R(%)	Iso-therm (R ²)	Kinet-ics (R ²)	Dose (g)	Time (min)	pH	Temp (°C)	Crys-tal-lite size (nm)	Sur-face area (m ² /g)	Reference
Bi ₁₂ CoO ₂₀ /ZnO	50	99%	Temkin	PSO	0.25	30	7	35			Madji et al., [69]
Cr@TiO ₂	50	383.84	Freundlich	PSO		1440	9	25			Zango et al., [137]
ZnO-Biochar	100	90.30%	Langmuir	PSO	0.05	120	5.5				Mudassar et al., [83]
org-NiONPs	20	56.24 mg/g	Langmuir (0.97)		0.025	120	5.8		6.71	128.3	Mahmoud et al., [72]
ppt-NiONPs	20	21.54 mg/g	Langmuir (0.93)		0.025	120	5.8		10	72.5	Mahmoud et al., [72]
NiONPs powder	100	65%	Freundlich	PSO		120	6	25			Abd El fatah and Ossman, [1]
CeO ₂ NPs	5-125	5.0 mg/g	Langmuir (0.999)	PSO (0.999)		720	6.8	30	> 10		Wang et al., [129]
MnFe ₂ O ₄ NPs		488 mg/g	Langmuir				5				Kumar et al., [59]
CoFe ₂ O ₄ NPs	100	32.11 mg/g	Langmuir (0.988)	PSO (0.967)	0.025				9.2	93.7	Culita et al., [27]
CoFe ₂ O ₄ NPs		63.1 mg/g			0.005		12				Al Yaqoob et al., [11]
SnO ₂ NPs	5	98.13 mg/g	Langmuir (0.985)		0.05	180	5	40	13	145	Rehman et al., [99]
TiO ₂ NPs	5	90.13 mg/g	Langmuir (0.927)		0.05	180	5	40	34	120	Rehman et al., [99]
TiO ₂ NPs	500	100%					5	25	8.3	185.5	Engates and Shipley, [35]
Flowerlike MgONPs	100	1980 mg/g	Langmuir		0.01		7			72	Cao et al., [21]
MgONPs	10	94.78%	Langmuir (0.96)		0.8	280	9			103.5	Dargahi et al., [28]

synthesize CNTs of high quality and purity. Chemical vapour deposition prefers a temperature less than 800 °C and an easy controlling design to synthesize high yield and pure CNTs [92]. Fig. 3 shows the walls of SWCNTs and MWCNTs.

Pristine CNTs are unique because of their structural stability, hydrophobicity, and high adsorption properties [125]. Pristine CNTs are pure carbon nanotubes without the presence of any chemical agent. In contrast, functionalized CNTs are CNTs that have been enhanced using a reagent. CNTs can be functionalized using oxidizing agents (such as nitric acid, sulfuric acid, potassium permanganate, hydrochloric acid, or hydrogen peroxide), organic compounds (such as cysteine, hydroxylamine, or hydrazine), polymers, and metal oxides [92]. The different possible fields of applications of CNTs are shown in Fig. 4.

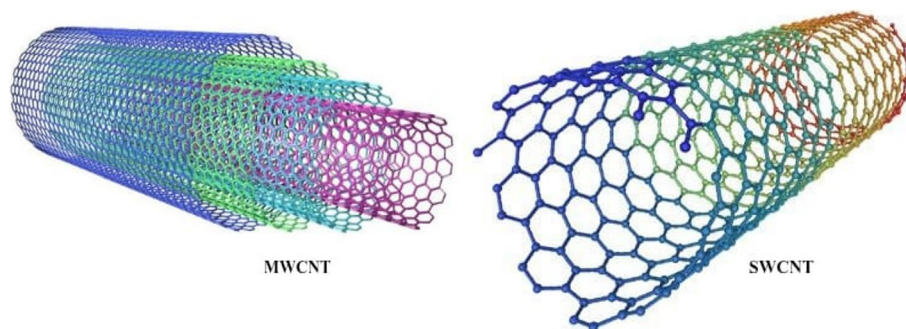


Fig. 3 Schematics of SWCNTs and MWCNTs [134]

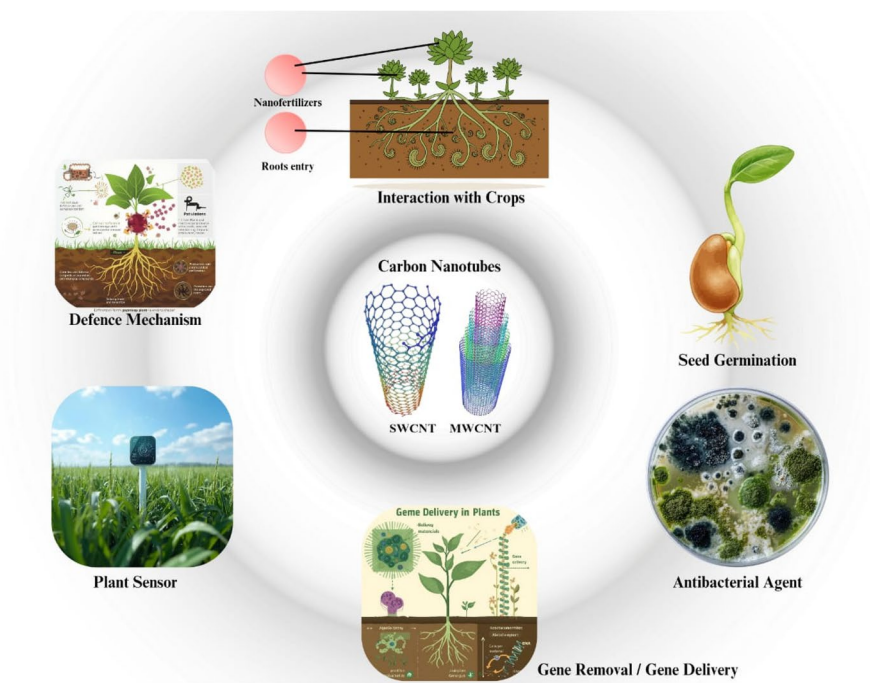


Fig. 4 Possible applications of CNTs

4.1 Multi-walled carbon nanotubes (MWCNTs)

Review of literature revealed that MWCNTs have been more extensively studied for Pb sequestration than SWCNTs due to their larger surface area and greater number of active sites for adsorption. These structural advantages, owing to the presence of multiple graphene layers and wider pore channels enhances adsorption capacity, enable functionalization, and improve stability during the sorption process [141]. The adsorption capacity of PCNTs typically fall within the range of about 10–215 mg/g, with MWCNTs consistently outperforming SWCNTs, therefore, the comparative synthesis focuses on MWCNTs.

The sequestration performance of the MWCNTs in the studies reviewed was observed to vary depending on surface area, particle size, surface modification, synthetic method, and experimental conditions. Removal efficiencies > 80% have been observed for commercially available MWCNTs.

The study also noted adsorption can also be dependent on other factors such as tube diameter and defect density, as even CNTs with similar particle sizes were noted to exhibit significantly different adsorption capacities. Abdel Salam et al., [2] observed a Pb adsorption capacity of 200 mg/g under optimized conditions using MWCNTs produced by Shenzhen Nanotech Port Co., Ltd., USA, with a diameter of 15–25 nm, while MWCNTs bought from Chengdu Organic Chemicals Co., Ltd., with a diameter of 10–20 nm, showed an adsorption capacity of 63.5 mg/g [138].

Similarly, a comparison of different-sized MWCNTs with particle sizes of 5–15 nm and 16–25 nm showed Pb adsorption capacities of 215.38 mg/g and 201.35 mg/g, respectively, under identical conditions, highlighting the effect of smaller particle sizes on adsorption capacities [33].

Reviewed studies also indicate substantial variability in adsorption performance based on the synthetic method. Mubarak et al., [82] reported that MWCNTs produced by microwave-assisted chemical vapor deposition achieved a Pb adsorption capacity of 104.2 mg/g, while Kucherova et al., [58] noted that MWCNTs synthesized using only the CVD method recorded a Pb adsorption capacity of 26.52 mg/g.

Gouda, [43] employed simple MWCNTs from Aldrich, Milwaukee, WI, USA, and reported adsorption efficiency > 96%. Abdel Salam et al., [2] recorded 99% adsorption efficiency for MWCNTs from Shenzhen Nanotech Port Co., Ltd. MWCNTs from MER Corp. were used to adsorb lead ions, and an adsorption efficiency of > 80% was observed [84]. On the other hand, Tehrani et al., [120] employed raw MWCNTs from Cheaptubes Inc., USA, and recorded a Pb adsorption capacity of 6.7 mg/g.

Table 5 captures the performance of some reported commercial and synthesized CNTs in the sequestration of Pb from aqueous solutions, while supplementary data S1 compares the performance against some MONPs.

5 Comparison between MONPs and PCNTs

The performance trade-offs between MONPs and CNTs were assessed based on the adsorption capacity, mechanism, kinetics, toxicity, cost, regeneration, and dispersion in water, and the summary findings are shown in Supplementary data S3.

5.1 Adsorption capacity and removal percentage

The comparative evaluation of the performance of MONPs and PCNTs reveal a consistent trend where material chemistry, rather than only surface area govern adsorption performance [66, 101]. The adsorption performance was observed to not completely depend on the available surface area, but also on the nature of surface chemistry and binding mechanism. Reported observations across reviewed studies indicate that both CNTs and MONPs demonstrate strong performance towards Pb, although the degree of performance varies depending on factors such as nanoparticles' composition, morphology, functionalization, experimental conditions, and synthetic routes [10, 57]. Although PCNTs are widely acknowledged for their high surface areas, however, this structural advantage does not directly translate to superior Pb removal ability. Rather, the presence, accessibility and reactivity of the surface groups have been reported as dominant factors influencing efficiency [133, 141].

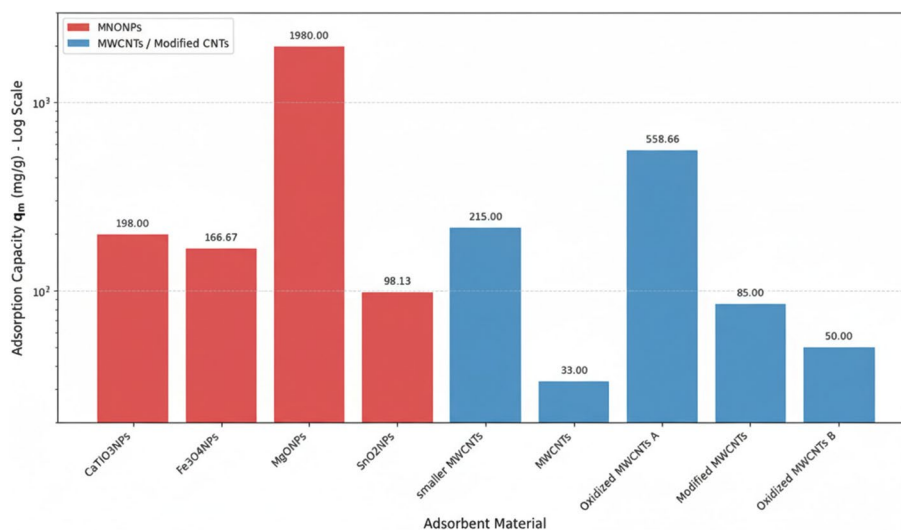
Among the MONPs reviewed, MgONPs exhibited some of the highest adsorption capacities reported, with values reaching as high as 1980 mg/g. Most of the MONPs

Table 5 Performance of P-CNTs in the sequestrations of Pb

CNTs	I.C (ppm)	q _m (mg/g) or R(%)	Iso-therm (R ²)	Kinet-ics (R ²)	Dose (g)	Time (min)	pH	Temp (°C)	Crys-tal-lite size (nm)	Sur-face area (m ² /g)	Reference
SWCNTs	20	< 30%	Lang-muir (0.999)	PSO (0.994)		90	5	20		400	Moradi and Zare, [78]
MWCNTs	100	200 mg/g	Lang-muir (0.993)	PFO (0.957)	0.03	90	4	25	15–25	40–600	Abdel Salam et al., [2]
MWCNTs	100	63.5 mg/g	Lang-muir	PSO		300	6.27	25	10–20	401.3	Zhang et al., [138]
P-CNTs	10	23.6 mg/g	Lang-muir (0.995)	PSO		40	7	25	15.9	169.7	Li et al., [63]
MWCNTs	1040	26.52 mg/g		PSO (0.994)		30			10–20		Kucherova et al., [58]
MWCNTs	10	104.2 mg/g	Freundlich (0.99)	PSO (0.999)		22.5	5		17–24	206	Mubarak et al., [82]
MWCNTs	50	60%				60	6	27	5–15	233	Aliasghar et al., [15]
Simple MWCNTs	0.1	> 96%				10	7		30–60	300	Gouda, [43]
MWCNTs	0.5	99%	Lang-muir	PSO (0.999)	0.13	5	7	60	20–40	69	Abdel Salam, [3]
MWCNTs	17	4.32 mg/g	Lang-muir	PSO (0.99)		90	7	20			Yu et al., [135]
raw-MW-CNTs	5	2.94 mg/g	Lang-muir (0.991)	PSO (0.971)		40	6.2	25	20–30		Vuković et al., [127]
MWCNTs	10	15.9 mg/g	Lang-muir (0.988)			1440	6	25			Ren et al., [100]
P-CNTs	100	27.3 mg/g	Lang-muir			180	5	25	14	261.22	Tian et al., [121]
CNTs	5	85%				40	5	25	20–30		Muataz et al., [81]
MWCNTs	~ 0.02	10.3 mg/g				30	6			600	Tuzen et al., [123]
MWCNTs	1	> 80%				3	5.3		2–15		Muñoz et al., [84]
Smaller-MWCNTs	62/66	215.38 mg/g	Lang-muir (0.999)	PSO (0.999)	0.03	60	5	50	5–15	130.6	Egbosiuba et al., [33]
Bigger-MWCNTs	62/66	201.35 mg/g	Lang-muir (0.999)	PSO (0.999)	0.03	60	5	50	16–25	215.38	Egbosiuba et al., [33]
MWCNTs	15	> 7.2 mg/g			0.025	360	7	25		162.16	Wang et al., [130]
MWCNTs	75	< 20 mg/g		PSO (0.9948)		40		25	5–30	400	Robati, [102]
Raw MWCNTs		6.7 mg/g	Freundlich (0.98)		0.01	45	5.7–6	25	20–30		Tehrani et al., [120]
CNTs		96.03%	Lang-muir (0.996)	PSO (0.999)	0.01	80	5	25			Kabbashi et al., [52]

Table 5 (continued)

CNTs	I.C (ppm)	q_m (mg/g) or R (%)	Iso- therm (R^2)	Kinet- ics (R^2)	Dose (g)	Time (min)	pH	Temp (°C)	Crys- tal- lite size (nm)	Sur- face area (m^2/g)	Reference
MWCNTs	10	33 mg/g	Lang- muir (0.9978)	PSO (0.999)	0.02	120	6	20			Zhao et al., [142]

**Fig. 5** Adsorption capacities of the MONPs against CNTs

frequently showed high adsorption capacities above 300 mg/g, removal efficiencies above 90% while some others, like MnO₂NPS and NiONPs, showed adsorption capacities below 20 mg/g.

On the other hand, some pristine CNTs reported showed high adsorption capacities within the range of 60–215 mg/g with removal efficiencies of above 90% reported with optimized experimental conditions. Some other CNTs have also been reported to show lower adsorption capacities of less than 15 mg/g, with the performance usually linked to factors such as pore sizes and synthetic routes.

Overall, although some CNTs demonstrated stable and high adsorption capacities under optimized experimental conditions, some of the MONPs comparatively outperform the CNTs by achieving much higher adsorption capacities under optimized experimental conditions. Fig. 5 and supplementary data S1 show the comparative performance of the MONPs against the pristine CNT.

5.2 Adsorption mechanism and kinetics

The adsorption mechanisms of both CNTs and MONPs differ as a result of the differences in their respective physicochemical properties.

From the reviewed literature, the sequestration of Pb by MONPs mostly occurs by ion exchange, surface complexation, and electrostatic interactions. Taqui et al., [119] described electrostatic attraction as the primary reason behind any adsorption, while Wang et al., [131], Karami [54], and Dargahi et al., [28] also mentioned electrostatic attraction as being the main driving force for the adsorption of metals in

various adsorbents, especially magnetite Nano-rods. On the other hand, with PCNTS, the adsorption of Pb is largely governed by π -electron interactions, electrostatic interactions, and surface bonding [133]. This means that functionalization would enable stronger coordination with Pb ions. Fig. 6 depicts different proposed sorption mechanisms of binding of Pb onto MONPs and CNTs.

A key insight from this study is that MONPs derive their superior performance from chemically active sites, particularly from surface hydroxyl groups that facilitate the electrostatic interactions and surface complexation with Pb [113]. The adsorption kinetics for MONPs and CNT have been reported to mostly follow the pseudo-second order model, indicating a chemisorption process. MONPs mostly exhibit rapid adsorption rates due to surface reactivity and abundant active sites. This means that equilibrium is usually achieved within a few minutes, depending on the particle size and dispersion [49]. CNTs also demonstrate relatively fast kinetics, although in comparison to MONPs, they may experience slower initial rates because diffusion into the nanotube pores can be rate-limiting [25].

5.3 Toxicity

Humans are exposed to nanomaterials in three ways: inhalation into the pulmonary system, absorption through the dermal system, and ingestion through the gastrointestinal system [112]. These NPs may concentrate in the body and cause certain health risks or problems.

MONPs generally exhibit variable degrees of toxicity depending on their composition. MONPs such as TiO_2 NPs have been reported to trigger injuries to the brain, liver, kidney, and spleen tissues [40]. Green-synthesized MONPs have been reported to have milder side effects than their chemically produced counterparts [53]. Some MONPs have been reported to be non-cytotoxic at low concentrations but toxic at high concentrations [75, 132], while some others have been noted to react with blood and bodily fluids, and affect the workings of the cardiovascular and the central nervous systems (CNS)

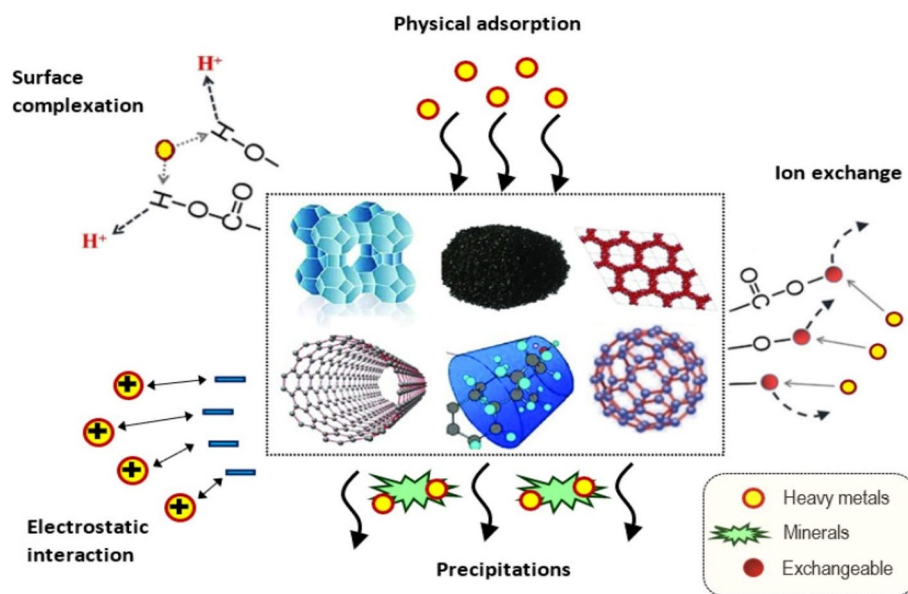


Fig. 6 Proposed mechanisms for adsorption [12]

[55]. Generally, the toxicity of MONPs has been reported to be mostly initiated through the reactive oxygen species, causing oxidative stress and DNA damage [50, 77].

On the other hand, CNTs have also been reported to pose significant toxicity, especially to the lungs and liver, with the potential to cause inflammation, oxidative stress, fibrosis, and cancer [32, 90, 105, 112]. Other studies have also reported adverse effects of CNTs on the CNS, kidney, eye, spleen, and the cardiovascular system [17]. CNTs have been reported to penetrate the seeds of plants, affecting the growth, and can be translocated even to the leaves [7, 56]. The toxicity of P-CNTs depends on diverse properties of the CNTs, such as length, aspect ratio, surface area, degree of aggregation, purity, concentration, dose, and other factors [23, 39].

Summarily, both MONPs and CNTs pose human toxicity risk, but whereas this risk stands unavoidable with CNTs and can only be avoided by wearing personal protective equipment while handling, the risk with MONPs can be avoided by choosing MONPs such as MgONPs, which have been reported to be of less toxicity and still afford high adsorption capacity, and employing green synthetic routes, which have also been recorded to reduce toxicity.

5.4 Cost

Chatterjee and Deopura, [22] reported low volume and high cost of production on a commercial scale as a demerit for P-CNTs. Nitodas and Karachalios, [85] reported thermal catalytic CVD as the most cost-effective method for synthesizing CNTs; given that synthesizing CNTs is an expensive endeavor, while other reports mentioned that CVD has the potential to drive down the cost of CNTs [127]. Laser ablation produces high-purity CNTs, but its use becomes disinteresting when its cost of production is analyzed [104]. Studies have reported that the application of CNTs since its discovery in 1991 has slowed down due to its high cost of production, with production cost reaching as high as \$10⁵ per ton [97, 138]. Zhang et al., [138].

In contrast, MONPs are known to be relatively cheap. Abdelrahman et al., [4], Mu et al., [80], and Nwoke et al., [87] reported that FeONPs, CuONPs, TiO₂NPs, ZnONPs, and MgONPs are relatively cheap adsorbents, with widespread applications reflecting the low production cost.

However, both CNTs and MONPs suffer from slow production processes, and the products are not specific in their properties without differences. The reason behind the problem of reproducibility is that the mechanism of nanoparticle formation has not been correctly and fully elucidated [92, 110].

5.5 Reusability and regeneration

Reusable adsorbents with good efficiency will always stand in preference over others as they offer practical and economic importance [23]. Regeneration studies demonstrate that both CNTs and MONPs can be reused effectively. MONPs generally maintain good performance across multiple adsorption-desorption cycles due to their structural stability and resistance to chemical degradation [101]. CNTs also exhibit excellent regeneration potential due to their mechanical strength and chemical inertness.

Overall, both MONPs and CNTs demonstrate strong reusability, although MONPs may be more cost-effective for repeated use.

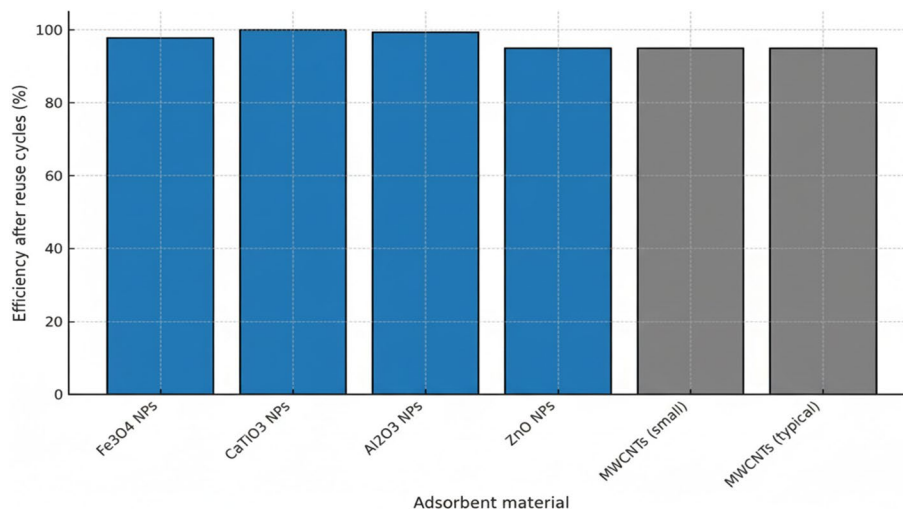


Fig. 7 Efficiency of the Nano-sorbents after multiple reuse cycles

Supplementary data S2 summarizes the reusable and regenerative abilities of MONPs and CNTs, while Fig. 7 shows the best-performing MONPs against the CNTs.

5.6 Dispersion in water

Dispersion behavior significantly affects sorption efficiency. Many MONPs disperse well in water owing to the presence of surface hydroxyl groups, although aggregation can occur at higher concentrations due to high surface energy and intermolecular forces [9].

CNTs, on the other hand, are inherently hydrophobic and tend to aggregate in water, necessitating the need for surface functionalization or the use of a dispersing agent [61, 90].

Therefore, MONPs demonstrate superior dispersion in water compared to CNTs. Supplementary data S3 summarizes the comparison of PCNTs and MONPs against the above indices.

5.7 Limitations

Despite the insights gleaned from this study, it is however, subject to several limitations. Most of the reported studies relied on batch-scale adsorption experiments that adopt synthetic solutions instead of real complex water samples. In addition, the reported performance efficiencies were the product of varied optimization conditions, and as such, no data reports the performance of these MONPs and the CNTs under the same experimental conditions. These differences in reporting standards and experimental conditions hinder direct comparisons. Furthermore, limited data exist on the environmental impact and the feasibility of large-scale water treatment applications for both adsorbents.

It is also noteworthy that most of the reported studies on the applications of P-CNTs are from the earlier years of CNT research, and this is likely due to the shift to the incorporation of functional groups and other forms of modification that promise to increase the performance of CNTs in heavy metal sequestration.

6 Conclusion

This review critically compared the performance of metal oxide nanoparticles (MONPs) and carbon nanotubes (CNTs) for Pb removal from aqueous systems. Both nanomaterials demonstrated high adsorption efficiency, rapid uptake kinetics, and good reusability, highlighting their potential for water treatment applications. However, MONPs generally exhibited higher adsorption capacities and faster adsorption rates, largely due to their abundant surface hydroxyl groups and ion-exchange sites. In addition, their simpler synthesis routes, better aqueous dispersibility, and lower production costs make them more attractive for large-scale implementation. In contrast, CNTs often require surface functionalization to improve dispersion and adsorption performance, increasing preparation complexity and cost. Although both adsorbent classes raise concerns regarding environmental and human health risks, the toxicity of MONPs can be moderated through careful selection of metal oxides, whereas toxicity concerns associated with CNTs remain more challenging to address. In addition, adsorption performance for both materials is strongly influenced by synthesis protocols, surface characteristics, and operating conditions, underscoring the need for improved reproducibility and standardization. Further studies should focus on standardized experimental conditions and large-scale water treatment applications, as this will be critical in advancing practical and sustainable nanotechnology solutions for Pb remediation in water treatment.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s11671-026-04782-5>.

Supplementary Material 1.

Authors' contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. The names of the authors are: Samuel Nnadi (samuelnnadi000@gmail.com), Ifeanyi Adolphus Ucheana (ucheara.ifeanyi@unn.edu.ng), Stanley Ugochukwu Nwoke (stanley.nwoke@unn.edu.ng).

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

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Consent for publication

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Competing interests

The authors declare no competing interests.

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